

Olist E-commerce Sales & Customer Satisfaction Analysis Report

1. Project Overview

Objective

The objective of this project is to analyze the Olist E-commerce dataset to identify the main factors affecting business revenue and customer satisfaction. The project includes data cleaning, database validation, exploratory data analysis, dashboard development, and business recommendations to support data-driven decision making.

2. Dataset Overview

The analysis is based on the Brazilian Olist E-commerce dataset, which contains online orders placed between September 2016 and October 2018.

The project uses the following datasets:

Orders

Order Items

Public

Payments

Reviews

Products

Customers

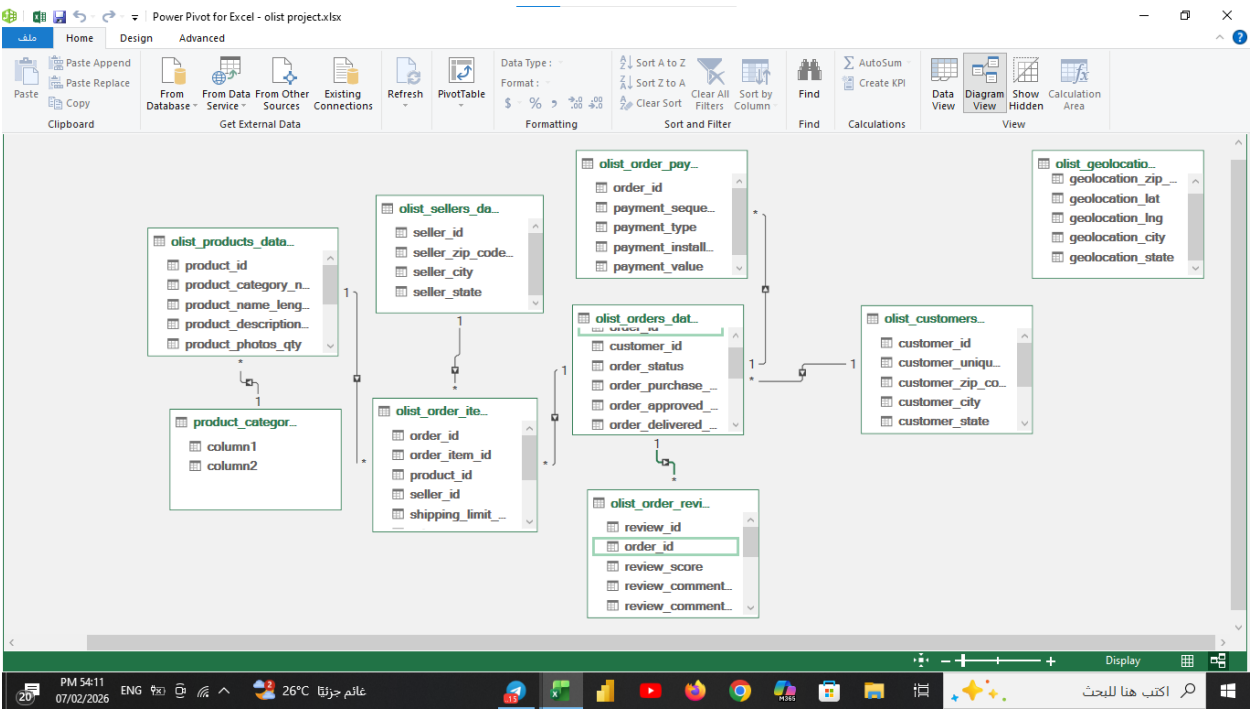
Sellers

Product Category Translation

Geolocation

3. Database Design

The following Entity Relationship Diagram (ERD) illustrates the database structure and relationships between all tables used throughout the analysis.



4. Data Cleaning & Data Quality Assessment

Before starting the analysis, the dataset was carefully examined to ensure data quality and consistency.

Data Cleaning Summary

The dataset covers orders placed between September 2016 and October 2018.

Some delivered orders contained missing approval or delivery timestamps. Since the correct values could not be verified, these records were retained.

Twenty-three orders contained inconsistent delivery dates where the customer delivery date occurred before the carrier delivery date.

Nine payment records contained a payment value equal to zero. These were associated with voucher and not_defined payment types and were retained.

Two credit-card payment records contained zero installments and were documented as anomalies.

One delivered order had no corresponding payment record.

Review titles and review messages contained missing values, while review scores were always available.

Four products had an invalid weight equal to zero, which was replaced with NULL.

Two products contained missing physical dimensions and were retained.

610 products had missing category information. These products represented only 1.3% of total revenue, so they were retained and documented.

Two missing category translations were manually added.

Missing geolocation coordinates were retained because they could not be recovered.

Database Integrity

To improve data integrity, the following constraints were implemented:

Primary Keys

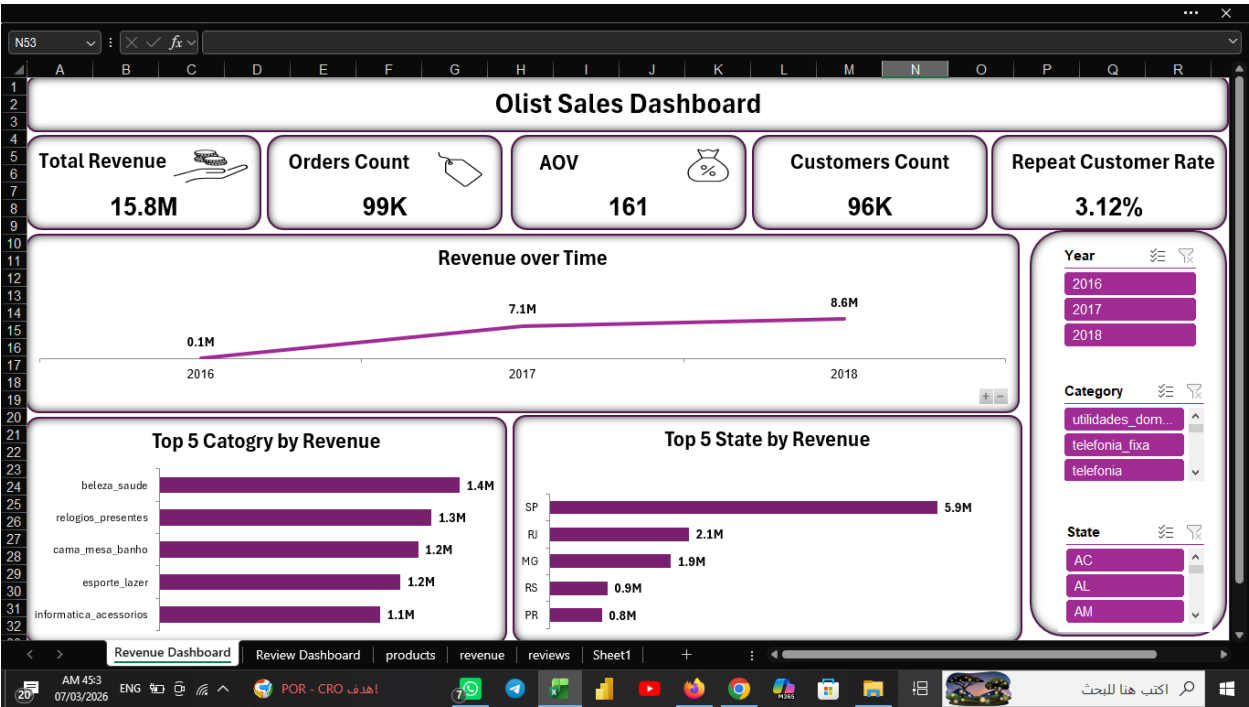
Foreign Keys

Check Constraints

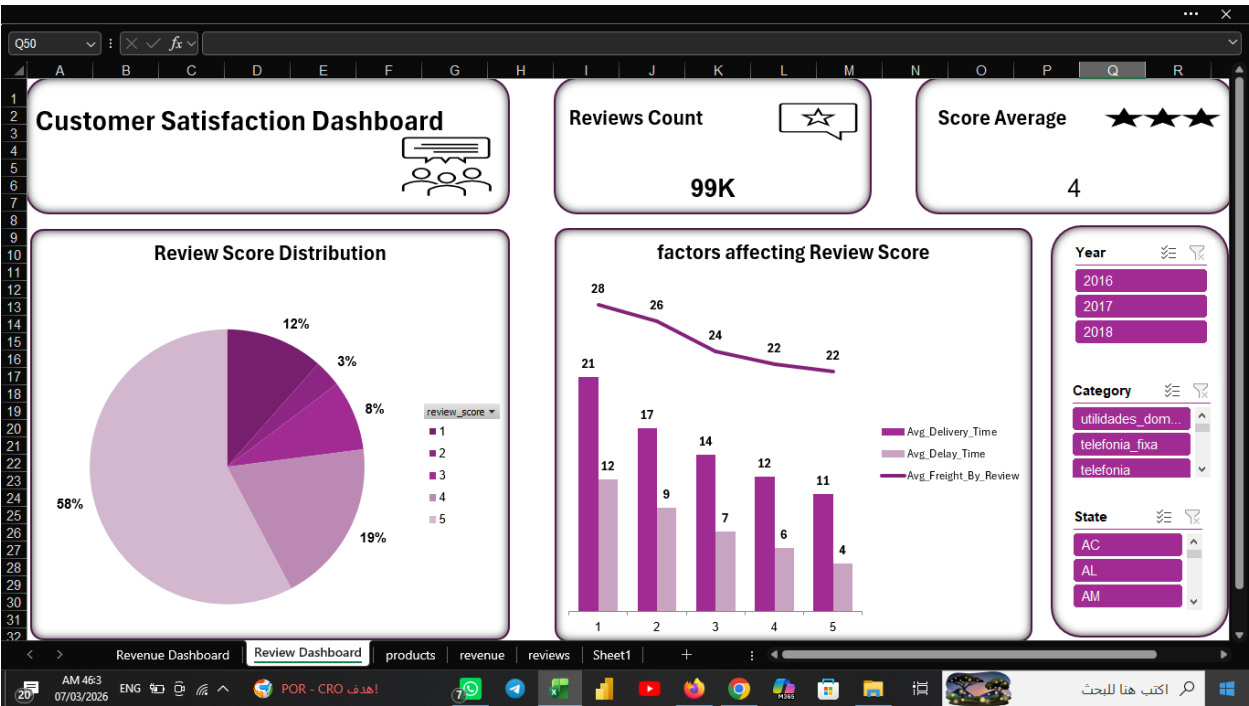
5. Dashboards

Two interactive dashboards were developed to summarize the business performance.

Revenue Dashboard



Customer Satisfaction Dashboard



6. Key Performance Indicators (KPIs)

The dashboards include the following KPIs:

Total Revenue

Total Orders

Total Customers

Average Order Value (AOV)

Average Review Score

Repeat Customer Rate

Cancelled Orders

7. Revenue Analysis

Revenue Trend

Insight

Revenue generally increased during the analysis period.

The unusually low revenue observed in September, November, and December 2016 and September 2018 resulted from incomplete data collection rather than poor business performance.

Recommendation

Exclude incomplete months when evaluating revenue trends to avoid misleading business conclusions.

Revenue by Product Category

Insight

The highest revenue was generated by:

Beauty & Health

Watches & Gifts

Bed, Bath & Table

Sports & Leisure

Computer Accessories

These categories contribute a large proportion of total business revenue.

Recommendation

Expand product assortment.

Maintain sufficient inventory.

Prioritize promotional campaigns for these categories.

Revenue by State

Insight

São Paulo (SP) generated significantly higher revenue than every other state.

Rio de Janeiro (RJ) and Minas Gerais (MG) ranked second and third respectively.

Recommendation

Consider expanding business operations in high-performing states.

Investigate low-performing states.

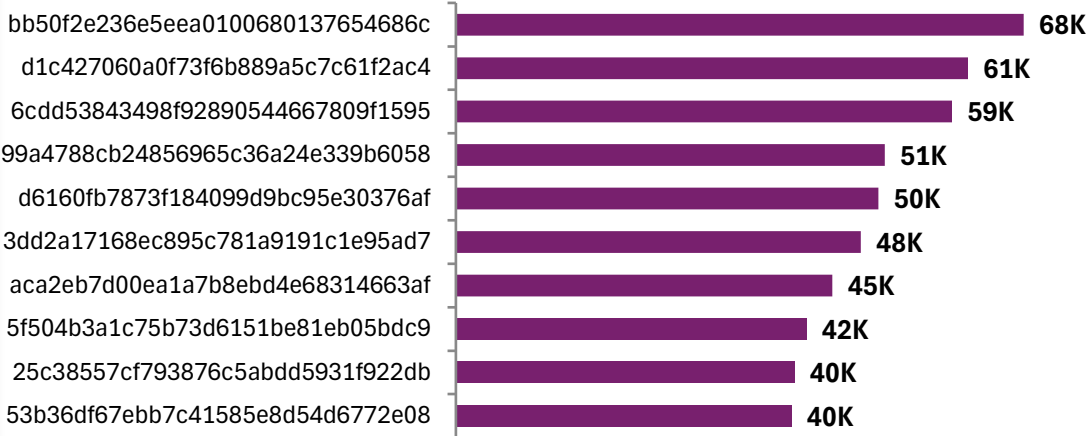
Launch targeted marketing campaigns where sales remain low.

Product Performance

The dataset provides only Product IDs rather than descriptive product names. Therefore, the product analysis is presented in this report instead of inside the dashboard.

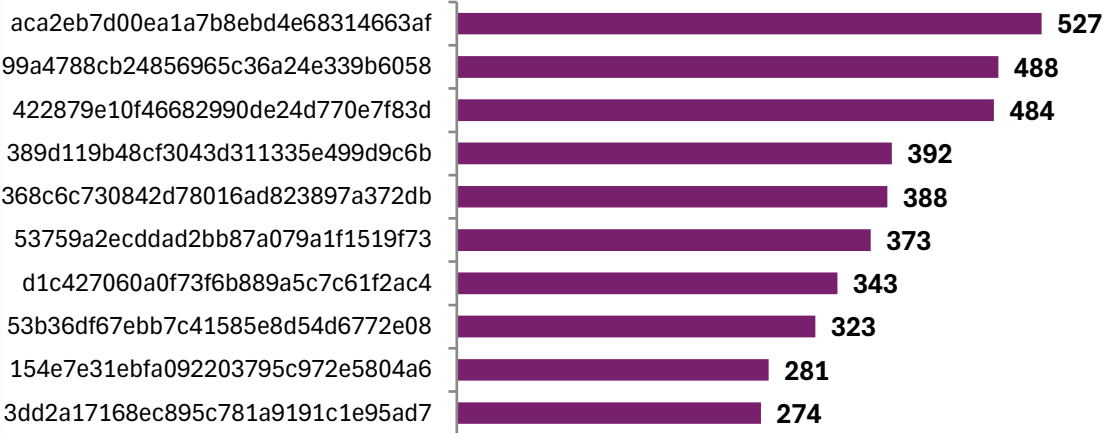
Top Products by Revenue

Top 10 products by Revenue



Top Products by Sales Quantity

Top 10 products by sales quantity



Several products consistently appear among both the highest-revenue and highest-selling products, indicating that they are major contributors to overall business performance.

Recommendation

Prioritize inventory management.

Ensure continuous stock availability.

Include these products in promotional campaigns.

Repeat Customer Analysis

Insight

Only 3.12% of customers made more than one purchase, indicating a very low customer retention rate.

Recommendation

Introduce loyalty programs.

Offer personalized discounts.

Launch post-purchase marketing campaigns.

Increase customer lifetime value through retention strategies.

8. Customer Satisfaction Analysis

Review Score Distribution

Insight

Most customers gave ratings between four and five stars, indicating generally high customer satisfaction.

Delivery Time

Insight

Longer delivery times are associated with lower review scores.

Recommendation

Improve logistics efficiency and reduce delivery time whenever possible.

Delivery Delay

Insight

Orders delivered after the estimated delivery date received noticeably lower customer ratings.

Recommendation

Reduce delivery delays through better shipment planning and logistics monitoring.

Freight Cost

Insight

Higher freight costs tend to be associated with lower review scores.

Recommendation

Optimize shipping costs whenever possible without affecting delivery quality.

Payment Type

Insight

The not_defined payment type received the lowest average review score.

However, only three orders used this payment type, making the sample statistically insignificant.

Recommendation

No business decision should be based on this payment type because of the extremely small sample size.

9. Order Characteristics

The analysis identified two important characteristics of the dataset:

Finding Result

Orders containing multiple product categories Yes

Orders containing multiple sellers Yes

These findings indicate that some orders include products from different categories and different sellers. Therefore, category-level and seller-level analyses were performed independently to avoid misleading conclusions.

10. Analyses with No Significant Relationship

Several analyses were performed but did not reveal meaningful business relationships.

These include:

Revenue vs Customer Satisfaction by State

Revenue vs Customer Satisfaction by Customer

Repeat Customers vs Review Score

Customer Satisfaction over Time

No statistically meaningful relationship was identified in these analyses.

11. Business Recommendations

Based on the analysis, the following recommendations are proposed:

Expand product assortment within high-performing categories.

Prioritize inventory management for top-performing products.

Expand operations in high-performing states after evaluating operational capacity.

Launch targeted marketing campaigns in underperforming states.

Improve delivery speed and reduce delivery delays.

Optimize freight costs where possible.

Introduce customer loyalty programs and personalized offers to improve customer retention.

Exclude incomplete periods from trend analysis when evaluating business performance.

12. Business Impact

This analysis identified the major drivers of revenue and customer satisfaction within the Olist e-commerce platform. High-performing product categories and states were identified as the primary growth opportunities, while delivery performance was found to have the strongest impact on customer satisfaction. In addition, the analysis revealed a very low customer retention rate, highlighting the need for customer loyalty initiatives. Implementing the proposed recommendations can help increase revenue, improve customer experience, and support more effective business decision-making.